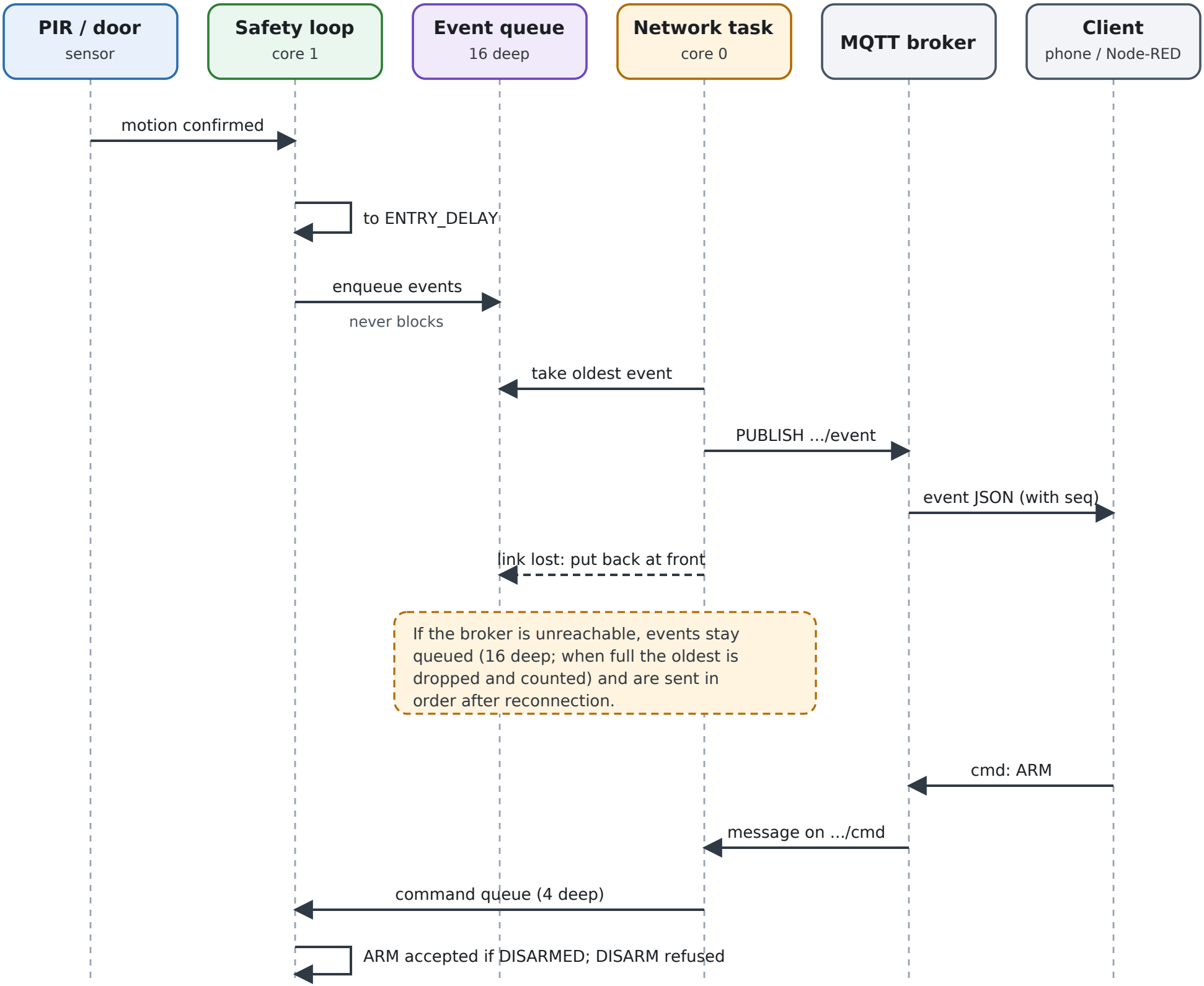
PIR / door
sensor
Safety loop
core 1
Event queue
16 deep
Network task
core 0
MQTT broker
Client
phone / Node-RED
motion confirmed
to ENTRY_DELAY
enqueue events
never blocks
take oldest event
PUBLISH .../event
event JSON (with seq)
link lost: put back at front
If the broker is unreachable, events stay queued (16 deep; when full the oldest is dropped and counted) and are sent in order after reconnection.
cmd: ARM
message on .../cmd
command queue (4 deep)
ARM accepted if DISARMED; DISARM refused